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Ripples of Curvature

in which gravitational waves carry to Earth
encoded symphonies of black holes colliding,
and physicists devise instruments
to monitor the waves
and decipher their symphonies

Symphonies

In the core of a far-off galaxy, a billion light-years from Earth and a billion years ago, there accumulated a dense agglomerate of gas and hundreds of millions of stars. The agglomerate gradually shrank, as one star after another was flung out and the remaining 100 million stars sank closer to the center. After 100 million years, the agglomerate had shrunk to several light-years in size, and small stars began, occasionally, to collide and coalesce, forming larger stars. The larger stars consumed their fuel and then imploded to form black holes, and pairs of holes, flying close to each other, occasionally were captured into orbit around each other.

Figure 10.1 shows an embedding diagram for one such *black-hole binary.* Each hole creates a deep pit (strong spacetime curvature) in the embedded surface, and as the holes encircle each other, the orbiting pits produce ripples of curvature that propagate outward with the speed of light. The ripples form a spiral in the fabric of spacetime around the binary, much like the spiraling pattern of water from a rapidly rotating lawn sprinkler. Just as each drop of water from the sprinkler flies nearly radially outward, so each bit of curvature flies nearly radially outward; and just as the outward flying drops together form a spiraling stream of water, so all the